NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Marks: 100	**ASSIGNMENT-5**	**Dead Line: 02-05-26**

Subject: Probability and Statistics **Sections: ALL**
Instructor: Mr. Hazber Samson, Miss Javeria, Miss Warda Date: 25-04-2026

Q1. (a) Consider the Poisson distribution with probability distribution function

$$f(x)=\frac{e^{-\lambda}\lambda^{x}}{x!},\ x=0,1,2,....$$

Find the maximum likelihood estimator of λ, based on a random sample of size *n*.

(b) Consider the Normal distribution with probability density function

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^{2}},\quad -\infty<x<\infty$$

Find the maximum likelihood estimator of μ *and* σ, based on a random sample of size *n*.

Q2. Suppose that $E(X_1)=\mu, Var(X_1)=7, E(X_2)=\mu, Var(X_2)=13, E(X_3)=\mu, Var(X_3)=20$ and consider the point estimates

$$\hat{\mu}_1=\frac{X_1}{3}+\frac{X_2}{3}+\frac{X_3}{3},\qquad \hat{\mu}_2=\frac{X_1}{4}+\frac{X_2}{3}+\frac{X_3}{5},\qquad \hat{\mu}_3=\frac{X_1}{6}+\frac{X_2}{3}+\frac{X_3}{4}+2$$

i. Calculate the bias of each point estimate. Is any one of them unbiased?

ii. Calculate the variance of each point estimate. Which one has the smallest variance?

iii. Calculate the mean square error of each point estimate. Which point estimate has the smallest mean square error when $\mu=3$?

Q3. The mean lifetime of electric light bulbs produced by a company has in the past been 1120 hours with a standard deviation of 125 hours. A sample of 100 electric bulbs recently chosen from a supply of newly produced bulbs showed a mean lifetime 0f 1070 hours. Company claimed that lifetime of bulbs has not changed. Test the claim at 5% level of significance using
(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

Q4. The Cholesterol levels of 8 women were measured, with the following results

$$3.1,\ 2.8,\ 1.5,\ 1.7,\ 2.4,\ 1.9,\ 3.3,\ 1.6.$$

Test at 5% level of significance, whether the sample has been drawn from a distribution with mean cholesterol level 3.1. using
(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

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Q5. A non-governmental organization wants to choose between two regions in a state to initiate a campaign for rainwater harvesting. A researcher claims that Region A receives lesser rainfall than Region B. To test the regions, the average rainfall is calculated for 60 days in each region. The mean rainfall in Region A is 700 millimeters. Assume the population standard deviation is 60 millimeters. The mean rainfall in Region B is 725 millimeters. Assume the population standard deviation is 66 millimeters. At $\alpha = 0.01$, can the organization support the researcher's claim? Use all the three approaches

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

Q6. Ms. Lisa is the budget director for Nexus Media Inc. She would like to compare the daily travel expenses for the sales staff and the audit staff. She collected the following sample information.

Sales ($)	131	135	146	165	136	142	
Audit ($)	130	102	129	143	149	120	139

At the 0.10 significance level, can she conclude that the mean daily expenses are greater for the sales staff than the audit staff? Check the claim using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*